

Data set	n	m	Ratio
Cancer	3.9k	217	1.14
Amazon	500k	10k	1.76
Muscle	220k	22k	2.47

Table 2: Timing the Hessian bound optimization scheme.

5. Software

We provide two software solutions in relation to the current paper. An R package, `msg1`, with a relatively simple interface to our multinomial and logistic sparse group lasso regression routines. In addition, a C++ template library, `sg1`, is provided. The `sg1` template library gives access to the generic sparse group lasso routines. The R package relies on this library. The `sg1` template library relies on several external libraries. We use the Armadillo C++ library [14] as our primary linear algebra engine. Armadillo is a C++ template library using expression template techniques to optimize the performance of matrix expressions. Furthermore we utilize several Boost libraries [15]. Boost is a collection of free peer-reviewed C++ libraries, many of which are template libraries. For an introduction to these libraries see for example [16]. Use of multiple processors for cross validation and subsampling is supported through OpenMP [17].

The `msg1` R package is available from CRAN. The `sg1` library is available upon request.

5.1. Run-time performance

Table 3 lists run-times of the current multinomial sparse group lasso implementation for three real data examples. For comparison, the `glmnet` uses 5.2s, 8.3s and 137.0s, respectively, to fit the lasso path for the three data sets in Table 3. The `glmnet` is a fast implementation of the coordinate descent algorithm for fitting generalized linear models with the lasso penalty or the elastic net penalty [10]. The `glmnet` cannot be used to fit models with group lasso or sparse group lasso penalty.

6. Conclusion

We developed an algorithm for solving the sparse group lasso optimization problem with a general convex loss function. Furthermore, convergence